

Developing a Hierarchically Structured Action Catalogue for an Elderly Care Robot through Participatory Design

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• Key words: Human Robot Interaction, Elderly care robot, Participatory design, Interaction design

1. Introduction

As companion robots become more and more common in our everyday life, the ability to perform contextually appropriate and emotionally meaningful actions becomes essential. Unlike humanoid robots, many service robots rely on constrained forms of expression, minimalist movement, gestures, or visual signals to communicate with intent. Designing such behaviors requires not only technological precision but also deep empathy and user-centered insight. Therefore, actions that can accurately reflect robot's behavior should be carefully extracted¹.

Lemmy, the focus of this study, is a non-humanoid, two-wheeled companion robot intended to support elderly individuals within home environments (Figure 1). Lemmy's design emphasizes subtle emotional expression and gentle behavioral cues through movement, screen-based animations, and sound. It is designed to provide reminders, emotional support, and assistive presence, promoting well-being without relying on overtly human-like traits.

This paper presents the development of Lemmy's action catalogue through a structured, participatory approach. Rather than engineering a predefined set of actions, we engaged designers in co-creating an intuitive and emotionally rich action catalogue grounded in user stories. The result of the overall process was not only a working foundation for Lemmy's interaction design but also a transferable methodology for deriving behavioral actions of socially assistive robots.

2. Literature Review

The design of expressive robotic behavior has gained growing attention across the domains of Human-Robot Interaction (HRI) ². Past studies emphasized the significance of emotional intelligence in HRI systems, particularly through non-verbal cues. Often, catalogued interaction



[Figure 1] Lemmy's (Companion Care Robot) 3D

Rendering to explain physical constraints

modalities in social robots mostly emphasized humanoid morphology³.

Researchers have also explored how user perception shifts when robots are perceived as beings that evolve over time⁴. In terms of designing robot actions, designers have used scenario enactment to develop proxemics-based motion strategies⁵. More recently, prior studies have focused on affective robot motion by combining user-driven annotation with dynamic motion libraries⁶.

Compared to these, our study emphasizes an action catalogue that prioritizes emotional subtlety and real-world relevance within the physical limitations of a non-humanoid, wheeled robot. It also leverages a co-design methodology aimed at producing actionable, user-aligned outputs that are both technically feasible and experientially valid.

3. Methodology

To generate the action catalogue, we conducted a participatory design workshop, involving four graduate and 2 postgraduate (total=6) students specializing in industrial design. Participants were grouped into teams of two.

Figure 2 illustrates the overall flow of the research process, including the workshop. At the beginning of each session of the workshop, participants were introduced to Lemmy

¹ Broadbent, E., Stafford, R., & MacDonald, B. (2009). Acceptance of healthcare robots for the older population: Review and future directions. *International Journal of Social Robotics*, 1(4), 319–330

² Breazeal, C. (2003). Emotion and sociable humanoid robots.

³ Fong, T., Nourbakhsh, I., & Dautenhahn, K. (2006). A survey of socially interactive robots.

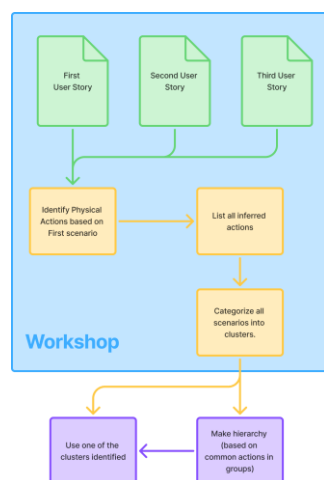
⁴ Lee, K. M., Park, N., & Song, H. (2012). Can a robot be perceived as a developing creature.

⁵ Takayama, L., & Pantofaru, C. (2009). Influences on proxemic behaviors in human-robot interaction.

⁶ Van Oers, C., Gijsenij, A., & Hung, H. (2020). Mapping motion to emotion: Motion design techniques for expressive social robots.

through visuals and a verbal overview. Importantly, Lemmy's physical constraints and communication capabilities (e.g., no limbs, screen-based expression, and differential wheel movement) were highlighted to ground participants in reality-based ideation.

Participants were then given three fictional but highly relatable daily-life user stories involving elderly users, such as preparing breakfast or searching for misplaced items. The task was to identify points where a robot could meaningfully intervene or support, then propose what the robot should do and how it should do it. These inferred robot actions were recorded, discussed, and iteratively clustered into conceptual categories by the participants themselves with guidance from the researcher. Clustering helped us identify key strategies on which robot actions can be divided.



[Figure 2] Flowchart of overall research process

After the workshop was complete, we divided actions by hierarchy based on commonality. This helped us identify the most intuitive robot actions. Furthermore, the actions present in only one group gave us the idea about unique actions that might be relevant only to the Lemmy robot.

4. Results

The workshop produced a rich and multifaceted understanding of what intuitive robot behavior might look like in an elderly home care context. Participants identified over 40 unique action suggestions, which were then grouped into broader themes, such as locomotion, ambient monitoring, emotional expression, and assistance.

Each group uniquely clustered their actions: group one clustered actions based on interaction with users (direct user interaction, indirect user interaction, ambient/time-initiated actions), the second group clustered on context (object-based actions, user-based actions, and schedule-based), and the last group clustered based on type of tasks.

Importantly, all three teams independently proposed actions that reflected social and emotional presence, rather than utilitarian efficiency. Behaviors like “pausing near the user until acknowledged,” “slightly rocking to express anticipation,” or “returning to dock after completing a task” demonstrate how design participants envisioned Lemmy as

not just functional but comforting. Furthermore, the hierarchical classification allowed us to identify actions most intuitively linked to a companion robot like Lemmy, offering valuable insights into which behaviors and functions should be prioritized for future development.

5. Discussion & Conclusion

The participatory process revealed varied interpretations of what a domestic robot should do in everyday contexts. While foundational actions like “approach,” “dock,” and “scan” were commonly identified, participants also emphasized the importance of ambient expressivity. This highlights the need for home robots to offer more than functional support; they should convey a passive social presence.

This research contributes to a methodologically grounded and user-informed action catalogue for Lemmy, an elderly care companion robot. Rooted in both literature and experience, the research contributes to the discourse around emotionally intelligent domestic robotics and sets the stage for real-world behavior design and testing. The action catalogue acts as a case study for future behavior-driven design in assistive robotics.

Future work will apply these insights toward variants of these actions in follow-up experiments. Additionally, we plan to develop an action catalogue toolkit based on the findings to accelerate research across diverse user groups working with domestic robots.

Acknowledgements

This research was supported by a research grant from the Ministry of Trade, Industry and Energy and the Korea Evaluation Institute of Industrial Technology (KEIT) in 2021 (20015056)

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